

GeoCalib: Calibrating Geometric Consistency for Reliable 3D Scene Graph Relations

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Abstract

3D Scene Graph predictors can assign high scores to relations that sound plausible but fail explicit geometry tests for the predicted object pair. We study this mismatch as a calibration problem: a source relation score ranks candidates but need not estimate predicate-conditioned geometric consistency. We introduce GeoCalib, a reliability framework for geometry-checkable relations. It separates predicate semantics T , predicate-independent same-pair geometry G , and source score Z ; fits a predicate-conditioned compatibility $C(T, G)$ without access to Z ; and combines Z and C only when re-ranking. We instantiate the framework with a family-calibrated product and a parameter-free rank-average fusion, and evaluate both with exact-label Recall@ K and verifier Violation@ K . On the Open3DSG source, the calibrated product improves Recall@100 from 0.5161 to 0.6047 while reducing Violation@100 from 0.1242 to 0.0341. At the primary $K = 100$ endpoint, both evaluated fusion rules satisfy the joint recall-violation criterion on SGFN. The calibrated product remains effective at smaller operating budgets, including $K = 50$ and $K = 10$, while factor-isolation, counterfactual, and object-pair identity controls reject score-copying and distance-only explanations. Across three predictors on a fixed 3DSSG target, the results support source-level generalization for geometry-identifiable relations; support/contact and transfer to a different dataset remain open.

Introduction

3D Scene Graphs turn reconstructed scenes into structured object-relation representations that can support spatial reasoning, embodied AI, scene alignment, visual grounding, and downstream decision making (Wald et al. 2020; Sarkar et al. 2023; Xie, Pagani, and Stricker 2024; Liu et al. 2026). For these uses, relation prediction must be supported by the reconstructed scene, not only plausible at the category or language level. A graph edge such as `standing on`, `next to`, or `higher than` is useful only if it describes the actual subject-object pair in 3D.

We study a failure mode in which relation predictors rank plausible labels that are inconsistent under explicit tests of

the actual object-pair geometry. A relation can sound plausible for two object categories while being contradicted by their contact evidence, distance, or vertical arrangement in the reconstructed scene. This mismatch matters more as 3D Scene Graphs become interfaces for language-facing, open-vocabulary, and embodied systems (Koch et al. 2024b; Hou et al. 2025; Zhang et al. 2025; Wang et al. 2025; Saxena and Chiun 2025; Yu et al. 2025; Madhavaram et al. 2026).

The failure is not simply that existing 3DSSG methods ignore geometry. Prior work already uses edge features, point-cloud structure, spatial knowledge, multimodal semantics, and geometric fusion (Zhang et al. 2021; Feng et al. 2023; Wang et al. 2023; Koch et al. 2024a; Chen et al. 2024). The gap is more specific: a source relation score—even when its predictor already uses geometry—is not necessarily an explicit estimate of whether the same subject-object pair satisfies the predicate in 3D. This makes the problem an AI reliability issue, not only a 3DSSG implementation detail: language-facing and embodied systems can inherit a confident symbolic relation that contradicts the underlying world state.

The resulting design requirement is factor isolation. GeoCalib treats an existing predictor as a relation source, standardizes its outputs into identity-preserving prediction rows, joins object-pair geometry evidence by subject and object instance identifiers, and separates predicate/family semantics T , predicate-independent same-pair geometry G , and source score Z . A source-independent compatibility factor $C = P(y^{cal} = 1 \mid T, G)$ is calibrated without Z ; the source score enters only in the final ranking $F(Z, C)$. This separation makes the mechanism falsifiable: wrong predicates, wrong pairs, and shuffled geometry should destroy compatibility while source confidence remains unchanged. A family-calibrated product preserves source-score magnitudes while penalizing family-conditioned geometry risk; rank-average combines semantic and geometry percentiles without assuming shared source scales. Pooled compatibility calibration is an ablation, Reciprocal Rank Fusion is a strong comparison, and hard-rule filtering is a diagnostic rather than the proposed operating point. Figure 1 summarizes this path.

We evaluate support/contact, proximity, and relative-vertical relations because their constraints are identifiable from reconstructed ordered-pair geometry. Open3DSG supplies the main open-vocabulary case study, VL-SAT a con-

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GeoCalib: Factor-Isolated Reliability for 3D Scene Graph Relations

The source proposes a plausible relation; same-pair geometry independently tests whether the predicate is supported.

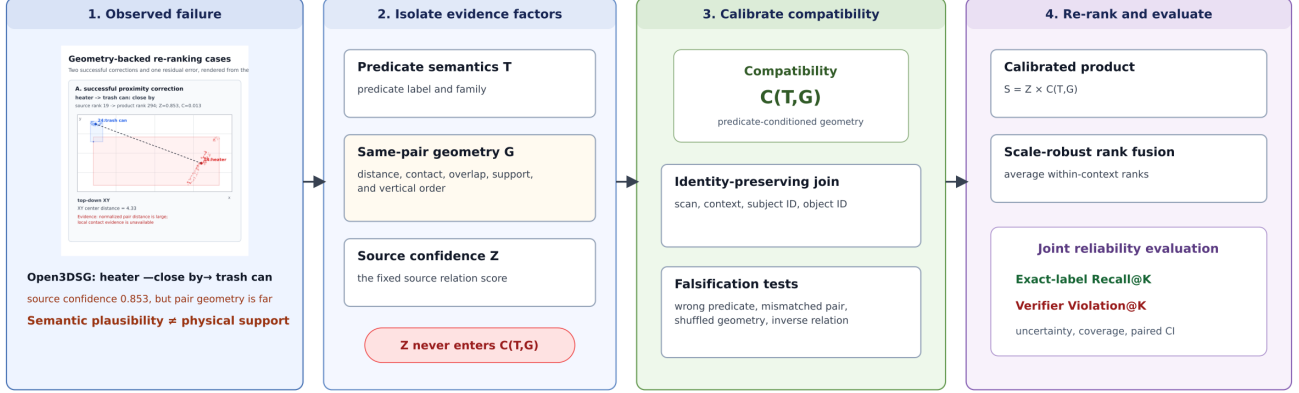


Figure 1: **Failure mechanism and GeoCalib overview.** A high-scoring Open3DSG relation is contradicted by the actual ordered-pair geometry. GeoCalib preserves pair identity, isolates predicate semantics T , raw geometry G , and source confidence Z , estimates $C(T, G)$ without Z , and combines the factors only for re-ranking. Falsification controls test whether the recall–violation trade-off depends on the correct predicate and object pair.

trolled closed-set anchor, and SGFN an additional source evaluation (Wang et al. 2023; Koch et al. 2024b). The evaluation reports exact-label recall together with verifier violation, uncertainty, coverage, paired confidence intervals, and family composition. ReplicaSSG/FROSS provides a cross-dataset transfer stress test and development diagnostic for source-scale and ranking-transfer failures.

Our contributions are threefold:

1. We identify and measure a post-source failure: a semantically confident relation can contradict the reconstructed geometry of its actual ordered object pair. We expose it through a joint exact-label recall, verifier-violation, uncertainty, and coverage contract.
2. We introduce a factor-isolated reliability layer that estimates predicate-conditioned compatibility $C(T, G)$ under $Z \notin C$ and combines it with source confidence only during re-ranking. Identity, wrong-predicate, and transformation controls make the mechanism falsifiable.
3. We establish source-level generalization across VL-SAT, Open3DSG, and SGFN under one geometry-identifiable 3DSSG target, and expose family heterogeneity and cross-dataset transfer failures rather than averaging them away.

The calibrated product improves the recall–violation operating point on the failure-exposing Open3DSG source, preserves a near-saturated VL-SAT anchor, and satisfies the primary joint criterion on SGFN. These results establish a reliability layer for geometry-checkable relations rather than a new relation generator.

Related Work

3D Scene Graph relation prediction. 3D Scene Graphs organize indoor scenes as object-relation structures grounded in 3D space (Armeni et al. 2019). The 3RScan/3DSSG setting established predicate recall for 3D relation prediction (Wald

et al. 2020), while incremental and online systems study how scene graphs can be constructed from RGB-D streams or spatial perception stacks (Wu et al. 2021; Hughes, Chang, and Carlone 2022). Recent predictors improve closed-set predicates using edge reasoning, point-cloud features, multi-modal cues, spatial knowledge, self-supervised pretraining, object-centric features, and edge-centric relational reasoning (Zhang et al. 2021; Feng et al. 2023; Wang et al. 2023; Koch et al. 2024a; Heo et al. 2025; Ma et al. 2026). These methods already use geometry; GeoCalib instead targets the post-source reliability gap that a high semantic relation score need not be calibrated to physical validity for the same object pair.

Open-vocabulary 3D graphs. Open-vocabulary 3D perception provides queryable object and region features (Peng et al. 2023; Takmaz et al. 2023); graph systems then connect such features to compact scene representations for querying, planning, navigation, online mapping, and language interaction (Gu et al. 2024; Werby et al. 2024; Maggio et al. 2024). Recent 3D scene graph work extends this line toward open-vocabulary objects, open-set relations, VLM features, online graph generation, and functional relations (Koch et al. 2024b; Chen et al. 2024; Hou et al. 2025; Zhang et al. 2025; Wang et al. 2025). This line raises the reliability requirement: when relation edges are language-facing scene representations, plausible text labels must still refer to object pairs that satisfy the corresponding spatial relation.

Other recent systems further connect VLMs, zero-shot graph construction, graph-mediated LLM reasoning, retrieval augmentation, and hallucination mitigation (Saxena and Chiun 2025; Madhavaram et al. 2026; Yu et al. 2025; Zemskova and Yudin 2025; Ogunleye, Abdelrahman, and Lourentzou 2026). We cite them as trend and boundary evidence, not direct GeoCalib baselines.

Geometry-aware relation evidence. A broad set of 3D reasoning methods already uses geometry: edge features, support/contact patterns, affordance geometry, metric distance, topology, and semantic-geometric fusion (Zhang et al. 2021; Feng et al. 2023; Huang et al. 2025; Shao et al. 2025; Xie, Pagani, and Stricker 2024). Scene graph alignment and registration methods further demonstrate that graph structure and geometric consistency matter when scene graphs are reused for matching, alignment, and downstream geometric tasks (Sarkar et al. 2023; Xie, Pagani, and Stricker 2024).

RelWitness uses visual-geometric witnesses, calibrated witness quality, and witness-consistent decoding for open-vocabulary 3D scene graph generation (Nguyen et al. 2026). Statistical confidence rescoring refines 3D scene graph predictions from multi-view images using semantic masks, neighboring relations, and statistical priors (Yeo, Li, and Lee 2025). GeoCalib therefore does not claim novelty from visual-geometric evidence, calibration, or rescoring alone. It studies a different post-source contract: identity-preserving geometry joins, recall/violation operating points, and controls over existing relation-source outputs.

Recent reliability-adjacent systems sharpen this boundary. SGFormer++ uses a spatial feature adapter inside an incremental predicate predictor (Qi et al. 2026); RelGraphOV prunes geometrically implausible relations while learning an open-vocabulary scene representation (Chen et al. 2026); and PUF performs uncertainty-aware semantic-spatial fusion for online node association and evidence accumulation (Yang et al. 2026). In contrast, GeoCalib leaves each source model fixed and audits already-produced relation candidates through a source-excluded, same-pair compatibility factor. The supplement compares these contracts directly.

Reliability and calibration. Standard R@K asks whether the correct predicate appears near the top, not whether the predicted relation is physically valid in the reconstructed scene. Confidence calibration asks whether model scores reflect empirical correctness likelihood (Guo et al. 2017), and selective prediction makes risk and coverage explicit (Geifman and El-Yaniv 2017). GeoCalib instantiates this reliability view for geometry-checkable 3DSSG families by standardizing prediction rows, joining same-pair 3D evidence, estimating compatibility, and reporting recall together with violation.

Problem Formulation

Let a relation source produce a set of candidate relation rows for a 3D scene. Each row is written as

$$r_i = (\text{scan}_i, \text{context}_i, s_i, o_i, p_i, a_i, Z_i), \quad (1)$$

where s_i and o_i are subject and object instance identifiers, p_i is the predicate label, a_i is the mapped predicate family, and Z_i is the source relation score. The tuple $(\text{scan}_i, \text{context}_i, s_i, o_i)$ is the identity key used to preserve the object pair across prediction, geometry, and ground-truth joins. We expose predicate semantics separately:

$$T_i = (p_i, a_i), \quad (2)$$

while withholding Z_i from geometry compatibility calibration.

For each prediction row, we join raw object-pair geometry evidence

$$G_i = G(\text{scan}_i, \text{context}_i, s_i, o_i). \quad (3)$$

The raw evidence is predicate-independent and includes oriented bounding-box features, local contact evidence, pair distance and scale, vertical ordering, and observability indicators. GeoCalib evaluates only geometry-checkable relation families:

$$\mathcal{A} = \{\text{support/contact, proximity, vertical order}\}. \quad (4)$$

Rows outside these families are excluded from the measured violation claim.

The geometry verifier assigns a row-level status $v_i \in \{\text{satisfied, uncertain, violated}\}$. The calibrator learns compatibility against a constructed calibration target:

$$C_i = P(y_i^{\text{cal}} = 1 \mid T_i, G_i), \quad Z_i \notin \text{inputs}(C_i). \quad (5)$$

Here y^{cal} contains training ground-truth positives and high-margin counterfactual negatives, not direct human validity. We denote family-conditioned and pooled implementations by C^{family} and C^{pool} . Predicate-aligned transformations are $T_i \times G_i$ interactions rather than raw geometry; neither Z_i nor source identity enters the calibrator. Final scores use $S_i = F(Z_i, C_i)$, with product and rank-average as soft instantiations rather than proofs or a universal formula.

Let $L_K(c)$ be the top- K measured-family rows for context c and Y_c the exact-label ground-truth set. We report

$$\text{R@K} = \frac{\sum_c |L_K(c) \cap Y_c|}{\sum_c |Y_c|}. \quad (6)$$

Family mapping determines only the geometry-checkable scope; it never relaxes predicate-label matching. Let D_K contain the selected rows whose verifier status is satisfied, uncertain, or violated. The primary diagnostic is

$$\text{V@K}_{\text{all}} = \frac{1}{|D_K|} \sum_{(c,i) \in D_K} \mathbf{1}[v_i = \text{violated}]. \quad (7)$$

Because uncertain rows lower this ratio without being declared satisfied, we also report decidable-only violation, uncertainty rate, and a pessimistic bound that counts every uncertain row as violated. Rows without a supported verifier status affect status coverage but no violation denominator. Recall, violation, uncertainty, and coverage are interpreted jointly.

Method

GeoCalib is a reliability layer for relation predictions, not a replacement 3D Scene Graph generator. It preserves candidate identity, measures predicate-conditioned geometry compatibility independently of the source relation score, and combines the two signals only at re-ranking time. Figure 1 gives the complete pipeline.

Relation Rows and Geometry Evidence

We convert each source prediction to a common row

$$(\text{scan}, \text{context}, s, o, p, a, Z), \quad (8)$$

where s and o are ordered instance identifiers, p is the predicate, a its relation family, and Z the original source relation score. The ordered key $(\text{scan}, \text{context}, s, o)$ joins each candidate to the geometry of the same object pair. It prevents category-level plausibility or unordered pair statistics from substituting for instance-level evidence.

The raw geometry vector G is predicate-independent. It contains pair distance and scale, centroid and extent differences, oriented-box overlap, and point-level contact/support evidence when observable. Predicate semantics are represented by $T = (p, a)$. We measure three families whose constraints can be tested from the reconstruction: support/contact, proximity, and relative vertical order. Family-specific transformations interpret G for T —for example, the sign of vertical displacement for higher than versus lower than—but neither source score nor source identity enters the geometry representation. Insufficient evidence is retained as uncertain rather than forced into a valid/invalid label.

Compatibility Calibration and Re-ranking

For each family a , we fit a logistic compatibility calibrator

$$C_i = \sigma(w_{a_i}^\top \Phi(T_i, G_i)) \approx P(y_i^{\text{cal}} = 1 \mid T_i, G_i), \quad (9)$$

$$Z_i \notin \text{inputs}(C_i),$$

with $\Phi = (\phi_T, \phi_G, \phi_\times)$. The interaction block $\phi_\times$ contains explicit predicate–geometry transformations, while ϕ_G remains predicate-independent. The calibration target uses 2,565 train/dev ground-truth positives and 3,244 high-margin counterfactual negatives; it is a constructed compatibility target, not human physical validity. We standardize numeric features with training statistics and minimize binary cross-entropy with $L_2 = 10^{-4}$ for 800 fixed gradient steps (learning rate 0.2). The three family models use 22–24 features and no source score or source identity. Training/dev rows, feature groups, and held-out calibration diagnostics are detailed in the supplement.

Our first instantiation preserves score magnitudes through a family-calibrated product,

$$S_i^{\text{prod}} = Z_i C_i. \quad (10)$$

Equivalently, $\log S_i^{\text{prod}} = \log Z_i + \log C_i$ applies a soft compatibility penalty with fixed unit weight. Our second instantiation avoids cross-source scale assumptions. Let q_i^Z and q_i^C be descending within-context percentile ranks; then

$$S_i^{\text{rank}} = \frac{1}{2}(q_i^Z + q_i^C). \quad (11)$$

The equal weight is parameter-free. These are complementary framework instantiations rather than a claim that either fusion is universally optimal.

Table 1: **Shared evaluation scope.** The candidate universe and exact-label denominator are fixed across source conditions.

Item	Count
Scans	157
Relation contexts	548
Directed object pairs	36,808
All ground-truth relations	11,254
In-scope ground-truth relations	3,972

Comparisons and Falsification Controls

We compare against the original source relation score, a pooled-compatibility product $Z_i C_i^{\text{pool}}$, fixed-constant Reciprocal Rank Fusion, and a hard-rule filter. We additionally fit a source-supervised nonlinear rescorer on the 117-scan internal-development source. It consumes Z , the complete T/G factorization, and the two directed interactions through a two-hidden-unit ReLU MLP with a predicate-linear skip. Its 69 trainable parameters match the total 22+23+24 coefficients of the three family compatibility calibrators. This is an advantaged comparison: unlike GeoCalib, it uses exact source-specific labels and is not a source-independent replacement. The pooled model tests whether family conditioning is necessary; the hard-rule result is a diagnostic operating point rather than a proposed zero-violation solution. A compatibility-only control removes Z , and a generic distance heuristic tests whether calibration is needed at all. Shuffled-geometry and mismatched-pair controls break the identity join while preserving marginal geometry statistics. A separate factor audit evaluates T -only, true G -only, additive $T + G$, and interaction $T \times G$ models, together with wrong-predicate, endpoint-swap where defined, and vertical inverse-equivariance tests. These controls make the central claim falsifiable: the gain should depend on the correct predicate, ordered object pair, and geometry—not on score leakage or a generic distance prior.

Experimental Setup

Datasets and evaluation scope. Open3DSG is the main open-vocabulary source case study, VL-SAT is a controlled closed-set anchor, and SGFN provides an additional source evaluation. Together they test source-level generalization across predictors under one fixed, geometry-identifiable 3DSSG target. All use the same scope: 157 scans, 548 relation contexts, 36,808 directed object pairs, and 3,972 exact-label ground-truth relations from support/contact, proximity, and relative vertical families (Table 1). Ground-truth membership is used only when computing recall, never to filter or rank candidate relations.

Open3DSG uses the official non-averaged model and a documented preprocessing recovery that completes all 548 contexts; the unmodified public pipeline covers 533. This is a source-output reliability case study, not a leaderboard reproduction. Exact model identity, preprocessing changes, and superseded partial-coverage branches are reported in the supplement rather than used as main-text evidence. VL-SAT and

SGFN each provide 957,008 candidate rows; the Open3DSG route provides 695,916.

Conditions and training boundary. We report the original source relation score, the family-calibrated product, rank-average fusion, pooled-compatibility product, Reciprocal Rank Fusion, and hard-rule filtering. A 69-parameter nonlinear rescorer provides a parameter-matched, source-supervised comparison. Controls comprise compatibility-only, distance-only, shuffled-geometry, and mismatched-pair rankings. The compatibility model uses only T , predicate-independent G , and explicit $T \times G$ interactions; source score and source identity are excluded. All normalization, imputation, feature preprocessing, and calibration parameters are estimated from training data. For GeoCalib, an internal development split is used to select the configuration before final evaluation. The nonlinear comparison is the sole exception: it is trained on exact-label correctness from the 117 internal-development scans and evaluated without refitting on the disjoint 157 final-validation scans.

For SGFN, we use the same family set, $K = 100$ primary endpoint, 3,972-relation denominator, fusion rules, RRF comparison, and paired-bootstrap evaluation. No predictions are synthesized for missing model edges. ReplicaSSG/FROSS is a cross-dataset transfer stress test and development diagnostic; its initial failure and a bounded-fusion follow-up are reported only in the supplement.

Metrics and uncertainty. Recall@ K uses exact predicate labels within each relation context; it is not directly interchangeable with leaderboard R@ K under a different candidate grouping. Violation@ K is the fraction of selected, geometry-checkable rows classified as violated; uncertain rows are not labeled valid. We report the all-status definition, violation among decidable rows, uncertainty rate, pessimistic violation, and decidable/status coverage. We use $K \in \{5, 10, 20, 50, 100\}$. The primary endpoint is $K = 100$; $K = 50$ is the canonical secondary budget, and $K = 10$ tests top-ranked operational reliability. We use paired 1,000-resample scan/context bootstrap intervals with shared indices across conditions. Within-family and global-top- K family slices expose composition effects that may be hidden by the aggregate metric.

Reproducibility. All reported experiments run in a fixed containerized environment; executable commands, dependency versions, hardware, and artifact manifests accompany the submission rather than occupying the method description.

Results and Discussion

Cross-Source Recall–Violation Trade-off

Table 2 asks whether calibrated geometry consistency reduces violations without collapsing recall. The Open3DSG source exposes the failure most clearly: source-score ranking places many geometrically inconsistent edges at the top of the list, especially for the low- K diagnostic. GeoCalib changes that operating point rather than merely pruning edges. The family-calibrated product at $K = 100$ improves recall by 8.86 percentage points over the source baseline and reduces

violation by 9.01 points; at $K = 5$ and $K = 10$, it also raises recall while moving violation from the high-error source-score regime to the calibrated-compatibility regime. Rank-average obtains nearly identical recall with a higher V, while RRF attains the highest recall but leaves more violations. The pooled product is weaker than family conditioning on this tradeoff, and hard-rule filtering is a zero-violation diagnostic rather than the default method.

VL-SAT provides the controlled anchor. Its source ranking is already close to ceiling, so the expected room for recall improvement is small. GeoCalib still reduces violation consistently across the reported K grid and gives a positive R@100 shift. RRF is strongest at $K = 100$ on both recall and V, whereas rank-average sacrifices low- K recall. This source dependence is why we report all fusion rules rather than selecting a universal winner.

The lower-budget results sharpen the formula boundary. On SGFN, the calibrated product improves both metrics at $K = 10, 50$, and 100, whereas rank-average and RRF sacrifice recall at $K = 10$ and 50. Thus the claim that two evaluated fusions satisfy the joint criterion is specific to the primary $K = 100$ endpoint; persistence across smaller budgets is supported by the calibrated product, not by every instantiation.

A 1,000-resample paired context bootstrap supports the same reading: Open3DSG has positive recall and negative violation shifts across K , while near-ceiling VL-SAT has smaller stable violation reductions. Main tables retain the standard fixed-source top- K point estimates.

An uncertainty sensitivity check rules out a denominator artifact. At $K = 100$, the family product reduces decidable-only V by .0244/.1671/.0428 on VL-SAT/Open3DSG/SGFN and also lowers the uncertainty rate on every source. The pessimistic definition that counts every uncertain row as violated remains lower by .0480/.2526/.0586; all paired 95% CIs exclude zero. Full status counts, coverage, and intervals are reported in the supplement.

Source Generalization, Learned Rescoring, and Family Effects

No fusion dominates across the observed sources: the family product has lower V on Open3DSG, while rank fusion is stronger on VL-SAT at $K = 100$. On SGFN, the family product changes R/V from 0.9235/0.0630 to 0.9416/0.0381 (paired deltas 0.0181 [0.0134, 0.0233] and -0.0249 [$-0.0270, -0.0229$]). Rank-average reaches 0.9476/0.0277; versus product its recall CI crosses zero and V CI is negative. This supports multiple fusion forms, not product superiority.

The parameter-matched nonlinear comparison is stronger when source-specific labels are available. It reaches R/V of 0.5441/0.0120 at $K = 10$, 0.8681/0.0186 at $K = 50$, and 0.9466/0.0279 at $K = 100$. At $K = 100$, its recall advantage over the family product is not significant, but its V reduction is; at smaller budgets it improves both. This rules out formula optimality and shows the value of supervised source adaptation. It does not explain away the source-independent setting addressed by GeoCalib: the MLP consumes exact SGFN labels, while the proposed compatibility factor is fitted without

Recall-violation trajectories across ranking budgets

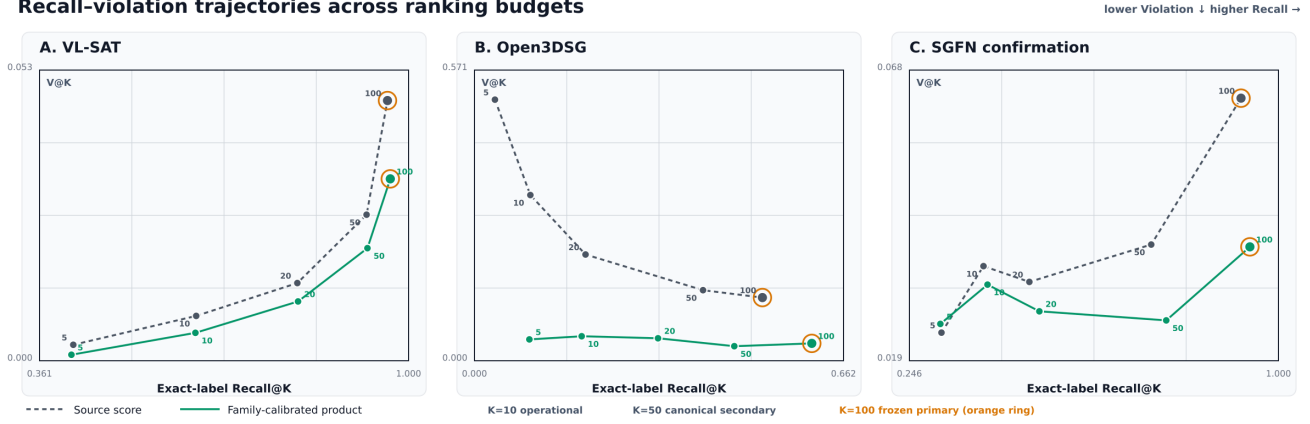


Figure 2: **Recall-violation trajectories.** Each line connects $K \in \{5, 10, 20, 50, 100\}$ for one fixed source and ranking rule. Rightward and downward movement is preferred. Orange rings mark the primary $K = 100$ endpoint; $K = 50$ is the canonical secondary budget and $K = 10$ the top-ranked operational diagnostic.

Geometry-backed re-ranking cases

Two successful corrections and one residual error, rendered from the actual ordered object-pair point clouds.



Figure 3: **Geometry-backed re-ranking cases.** Actual ordered-pair point clouds show successful proximity and support/contact demotions, plus a residual support error that remains at product rank 10 despite $C \approx 1$. Red denotes the subject and blue the object.

source identity, source scores, or source-specific correctness labels.

The aggregate effect is not family-uniform. On SGFN, the family product’s support/contact verifier V increases by 0.00450 (95% CI [0.00370, 0.00532]), proximity is unchanged, and relative-vertical relations supply most of the reduction. The same support/contact direction appears in the retrospective sources. We therefore claim aggregate scoped reliability and expose the family decomposition rather than claiming every-family improvement. The supplement reports within-family and actual global-top- K slices separately, so this conclusion does not depend on hiding a change in family composition.

Controls and Validity Checks

Table 3 rules out several simpler alternatives. Compatibility-only is not pooled re-ranking: it removes the source score, although it still contains T and $T \times G$. Distance-only is weaker, and shuffled/mismatched-pair controls show that object-pair identity is necessary. Thus GeoCalib is not merely ranking by a compatibility score, distance, or hard rules. A strict split audit adds true raw- G , T -only, additive, interaction, and counterfactual controls. Correct-predicate compatibility wins on 97.44% of 390 vertical ground-truth rows, endpoint/inverse-equivariance errors remain below 1.6%, and correct pairs exceed mismatched pairs by 0.423 over 3,961 rows. The audit does not imply that support/contact is solved.

The ground-truth/counterfactual check isolates the geometry signal from source re-ranking: ground-truth posi-

Table 2: **Main source results.** Recall uses the fixed 3,972-relation exact-label denominator; verifier V uses selected rows with a supported status. Product and rank-average are framework instantiations, RRF is a strong comparison, and hard-rule filtering is diagnostic.

Source	Condition	R@K					Violation@K				
		@5	@10	@20	@50	@100	@5	@10	@20	@50	@100
Open3DSG	Source score	0.0368	0.1002	0.1991	0.4096	0.5161	0.5131	0.3255	0.2088	0.1386	0.1242
Open3DSG	Family product	0.0984	0.1921	0.3291	0.4658	0.6047	0.0420	0.0482	0.0441	0.0286	0.0341
Open3DSG	Rank-average	0.1047	0.1939	0.3061	0.4733	0.6052	0.0369	0.0462	0.0440	0.0407	0.0532
Open3DSG	RRF	0.0984	0.1735	0.2769	0.4663	0.6196	0.0336	0.0411	0.0638	0.0946	0.0789
Open3DSG	Pooled product	0.0826	0.1581	0.2603	0.3975	0.5723	0.0628	0.0699	0.0654	0.0606	0.0811
Open3DSG	Hard-rule filter	0.0707	0.1314	0.2422	0.4295	0.5368	0.0000	0.0000	0.0000	0.0000	0.0000
VL-SAT	Source score	0.4194	0.6322	0.8074	0.9272	0.9635	0.0029	0.0082	0.0142	0.0268	0.0476
VL-SAT	Family product	0.4162	0.6309	0.8087	0.9288	0.9683	0.0011	0.0051	0.0109	0.0206	0.0333
VL-SAT	Rank-average	0.2142	0.3600	0.5279	0.7988	0.9597	0.0073	0.0109	0.0168	0.0199	0.0259
VL-SAT	RRF	0.2362	0.4134	0.6370	0.8920	0.9698	0.0073	0.0095	0.0124	0.0163	0.0251
VL-SAT	Pooled product	0.4154	0.6322	0.8107	0.9305	0.9688	0.0015	0.0071	0.0120	0.0229	0.0404
VL-SAT	Hard-rule filter	0.4197	0.6317	0.8074	0.9257	0.9627	0.0000	0.0000	0.0000	0.0000	0.0000

Table 3: **Falsification controls.** Compatibility-only removes the source score but still uses T and $T \times G$. Shuffled and mismatched-pair conditions break the identity-preserving geometry join.

Source	Condition	R@100	Violation@100	Interpretation
Open3DSG	Compatibility only	0.5116	0.0865	Removing Z does not recover the main tradeoff.
Open3DSG	Distance only	0.5038	0.1071	Generic distance is weaker than calibrated compatibility.
Open3DSG	Shuffled geometry	0.2543	0.1998	Breaking row-level geometry destroys reliability.
Open3DSG	Mismatched-pair geometry	0.2331	0.1985	Same-pair identity is necessary.
VL-SAT	Compatibility only	0.5184	0.0711	Removing Z loses substantial recall.
VL-SAT	Distance only	0.5554	0.0981	Distance is a poor proxy for relation reliability.
VL-SAT	Shuffled geometry	0.9494	0.0588	Violations increase when geometry identity is broken.
VL-SAT	Mismatched-pair geometry	0.9529	0.0601	Wrong object-pair evidence is worse than GeoCalib.

tives remain nonviolated at 0.9965, counterfactual negatives are nonsatisfied at 0.9673, and C^{pool} separates 7,944 rows with AUROC/AUPRC 0.9772/0.9729 and Brier 0.0543. This shows that soft compatibility is not identical to hard filtering, while remaining a verifier-derived diagnostic.

Qualitative Failures

In the 36-case Open3DSG queue, 25 relations are demoted by geometry-aware re-ranking. Figure 3 grounds two such distance/support contradictions and one residual error: the door-floor relation receives $C \approx 1$ and moves from rank 21 to 10 despite a verifier violation. Compatibility is therefore a soft factor, not a validity label.

Limitations

GeoCalib assumes a geometry-identifiable predicate and known object instances. We test support/contact, proximity, and vertical order; functional predicates remain out of scope, front/behind is frame-dependent, and left/right fails the defined controls. The task is relation reliability, not complete open-vocabulary graph generation.

Violation@ K is verifier-derived, and its geometric primitives overlap with the compatibility target. Counterfactual controls test the learned factor but do not replace an independent physical-validity reference. Moreover, support/contact

regresses, and the initial ReplicaSSG/FROSS transfer exposes source-score scale, ranking-displacement, geometry-representation, and ontology shift. ReplicaSSG is used as a development diagnostic rather than an unbiased external evaluation: bounded fusion preserves recall in an all-scene fit but fails the leave-one-scene-out recall guardrail. The evidence therefore establishes source-level generalization on one 3DSSG target, not uniform family or dataset-wide generality.

A source-supervised nonlinear rescorer also outperforms the parameter-free product when exact development labels are available. Our claim is thus source-independent factor isolation, not formula optimality. Future work should validate the metric independently, improve contact evidence, and test downstream decision benefits.

Conclusion

GeoCalib isolates source confidence from predicate-conditioned object-pair geometry. Across three predictors on a fixed 3DSSG target, this separation improves the aggregate recall-violation trade-off, and the calibrated product persists across smaller ranking budgets. GeoCalib contributes a source-independent reliability framework with explicit family and transfer limits, not a universally optimal fusion rule.

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